

SBT-DF203 — BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

LAB 4 — SMTP EMAIL TRAFFIC FORENSICS

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1. Executive Summary

This practical examines an authorised historical SMTP packet capture (smtp.pcap) to identify when email exchange occurred, identify client/server systems, reconstruct SMTP commands and message content, decode relevant Base64 values offline, and assess whether STARTTLS/TLS protected subsequent traffic. The original smtp.pcap is preserved unchanged in an original/ directory, and all evidence-dependent values in this report are populated directly from terminal execution and Wireshark/TShark analyses.

2. Scenario and Objectives

Scenario: Analyse a historical SMTP packet capture to determine when email exchange occurred, identify client/server systems, reconstruct SMTP commands/message content, decode relevant Base64 fields offline, and document network/encryption evidence safely.

- **Objectives:**

- Explain SMTP and common ports.
- Identify SMTP commands and response codes.
- Reconstruct the relevant SMTP TCP stream.
- Decode required Base64 training values offline while redacting sensitive values in the report.
- Extract message headers, client software indicators, IP addresses, TCP ports, and MAC addresses.
- Assess STARTTLS/TLS and state the evidential limitations caused by encryption.

3. Authorisation, Safety and Evidence Handling

Only the authorised SMTP training PCAP supplied for SBT-DF203 was used. No live, third-party, public, production, or wireless traffic was intercepted. No real credentials or sensitive emails were transmitted. The original capture file was preserved, and analysis was executed entirely on a verified working copy.

Requirement	Recorded action
Original evidence	smtp.pcap preserved unchanged
Working copy	~/SBT-DF203-Lab4/working/smtp.pcap
Authorised source	SBT-DF203 SMTP training resource
Analysis environment	Kali Linux 2024.1 / Wireshark 4.2.0 / TShark 4.2.0
Sensitive data handling	Credentials and unnecessary personal content redacted

4. Evidence Preservation and Hashing

Created a working directory and preserved the original before analysis.

```
mkdir -p ~/SBT-DF203-Lab4/{original,working,output,logs,screenshots}
cp smtp.pcap ~/SBT-DF203-Lab4/original/smtp.pcap
cp ~/SBT-DF203-Lab4/original/smtp.pcap ~/SBT-DF203-Lab4/working/smtp.pcap
sha256sum ~/SBT-DF203-Lab4/original/smtp.pcap
sha256sum ~/SBT-DF203-Lab4/working/smtp.pcap
cmp ~/SBT-DF203-Lab4/original/smtp.pcap ~/SBT-DF203-Lab4/working/smtp.pcap

capinfos ~/SBT-DF203-Lab4/working/smtp.pcap
```

Evidenc	Value
---------	-------

e item	
Original SHA- 256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
Working-copy SHA- 256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
Hash match	MATCH
File size	27 kB (26,866 bytes data size)
Packet count	60
Capture start	2009-10-05 02:06:07.492060
Capture end	2009-10-05 02:06:16.690444
Capture duration	9.198384 seconds
Link- layer type	Ethernet

```

(kali@kali)~$
$ mkdir -p ~/SBT-DF203-Lab4/{original,working,output,logs,screenshots} 66 \
cp ~/Downloads/smtp.pcap ~/SBT-DF203-Lab4/original/smtp.pcap 66 \
cp ~/SBT-DF203-Lab4/original/smtp.pcap ~/SBT-DF203-Lab4/working/smtp.pcap 66 \
sha256sum ~/SBT-DF203-Lab4/original/smtp.pcap 66 \
sha256sum ~/SBT-DF203-Lab4/working/smtp.pcap 66 \
cmp ~/SBT-DF203-Lab4/original/smtp.pcap ~/SBT-DF203-Lab4/working/smtp.pcap 66 \
capinfos ~/SBT-DF203-Lab4/working/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab /home/kali/SBT-DF203-Lab4/original/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab /home/kali/SBT-DF203-Lab4/working/smtp.pcap
File name: /home/kali/SBT-DF203-Lab4/working/smtp.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 60
File size: 27 kB
Data size: 26 kB
Capture duration: 9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time: 2009-10-05 02:06:16.690444
Data byte rate: 2,920 bytes/s
Data bit rate: 23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256: 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1: 3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
Encapsulation = Ethernet (1 - ether)
Capture length = 65535
Time precision = microseconds (6)
Time ticks per second = 1000000
Number of stat entries = 0
Number of packets = 60

```

Evidence Checkpoint 1: screenshot showing the original/working-copy hashes and capinfos output.

5. SMTP Overview and Common Ports

Port	Typical use	Security note
25	SMTP server-to-server / relay	May be plaintext unless STARTTLS is negotiated.
587	Message submission	Common submission port; often uses STARTTLS.
465	SMTP over implicit TLS	TLS is established before SMTP commands are exposed.

6. Lab Environment

Component	Recorded value
Operating system	Kali Linux 2024.1
Wireshark version	Wireshark 4.2.0

TShark version	TShark 4.2.0
PCAP	smtp.pcap
Analysis mode	Offline / historical PCAP
Analyst	IKPI GODWIN EDET

7. SMTP Conversation Inventory

Used TShark to identify TCP conversations and SMTP-related traffic.

```
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -q -z conv,tcp
```

```
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y smtp -T fields \
```

```
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e
```

```
tcp.dstport \
```

```
-e tcp.stream -e smtp.request.command -e smtp.response.code
```

```
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y 'tcp.port == 25 ||
```

```
tcp.port == 587 || tcp.port == 465' \
```

```
-T fields -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e
```

```
tcp.dstport -e tcp.stream
```

TCP stream	Client endpoint	Server endpoint	Ports	Packets	Observation
0	10.10.1.4:1470	74.53.140.153:25	1470 → 25	53	Single active TCP stream carrying SMTP traffic

```
(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -q -z conv,tcp
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y smtp -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
-e tcp.stream -e smtp.request.command -e smtp.response.code

TCP Conversations
Filter:<No Filter>

| Frames | Bytes | | Frames | Bytes | | Total | | Relative | | Duration |
|-----|-----| |-----|-----| |-----| |-----| |-----|
10.10.1.4:1470 ↔ 74.53.140.153:25 | 25 | 1,980 bytes | 28 | 22 kB | 53 | 24 kB | 0.036986000 | 7.5777

tshark: Some fields aren't valid:
smtp.request.command
```

Evidence Checkpoint 2: screenshot showing the SMTP conversation inventory.

8. Endpoint Identification

Role	IP address	TCP port	MAC address	Evidence packet(s)
SMTP client	10.10.1.4	1470	00:e0:1c:3c:17:c2	Frame 3, 5, 7 / Stream 0
SMTP server	74.53.140.153	25	00:1f:33:d9:81:60	Frame 4, 6 / Stream 0

```
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y smtp -T fields \
-e frame.number -e eth.src -e eth.dst -e ip.src -e ip.dst -e tcp.srcport -e
tcp.dstport -e tcp.stream
```

```
(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y "tcp.port == 25" -T fields \
-e frame.number -e eth.src -e ip.src -e tcp.srcport -e eth.dst -e ip.dst -e tcp.dstport | head -n 5
3      00:e0:1c:3c:17:c2      10.10.1.4      1470      00:1f:33:d9:81:60      74.53.140.153      25
4      00:1f:33:d9:81:60      74.53.140.153      25      00:e0:1c:3c:17:c2      10.10.1.4      1470
5      00:e0:1c:3c:17:c2      10.10.1.4      1470      00:1f:33:d9:81:60      74.53.140.153      25
6      00:1f:33:d9:81:60      74.53.140.153      25      00:e0:1c:3c:17:c2      10.10.1.4      1470
7      00:e0:1c:3c:17:c2      10.10.1.4      1470      00:1f:33:d9:81:60      74.53.140.153      25

(kali@kali)-[~]
$
```

Evidence Checkpoint 3: screenshot showing endpoint IPs, ports and MAC addresses.

9. 220 Service-Ready Response

```
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y 'smtp.response.code
== 220' -T fields \
```

-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e smtp.response.code -e smtp.response.parameter

Fram e	Time	Server IP	Client IP	TCP strea m	Cod e	Greeting text
6	05 Oct 2009 01:05:5 4 - 0500	74.53.140.1 53	10.10.1. 4	0	220	220- xc90.websitewelcome. com ESMTP Exim 4.69

```
(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y "smtp" -V | grep -E "Frame |Response: 220|Response code: 220"
Frame 6: 235 bytes on wire (1880 bits), 235 bytes captured (1880 bits)
  Frame Number: 6
  Frame Length: 235 bytes (1880 bits)
  [Frame is marked: False]
```

```
Session Actions Edit View Help
[Frame is ignored: False]
Response: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 \r\n
Frame 7: 63 bytes on wire (504 bits), 63 bytes captured (504 bits)
  Frame Number: 7
  Frame Length: 63 bytes (504 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
Frame 9: 191 bytes on wire (1528 bits), 191 bytes captured (1528 bits)
  Frame Number: 9
  Frame Length: 191 bytes (1528 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
Frame 10: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
  Frame Number: 10
  Frame Length: 66 bytes (528 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
Frame 11: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
  Frame Number: 11
  Frame Length: 72 bytes (576 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
Frame 12: 84 bytes on wire (672 bits), 84 bytes captured (672 bits)
  Frame Number: 12
  Frame Length: 84 bytes (672 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
Frame 13: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
  Frame Number: 13
  Frame Length: 72 bytes (576 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
Frame 14: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
  Frame Number: 14
  Frame Length: 72 bytes (576 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
Frame 15: 84 bytes on wire (672 bits), 84 bytes captured (672 bits)
  Frame Number: 15
  Frame Length: 84 bytes (672 bits)
```

Evidence Checkpoint 4: Wireshark screenshot clearly showing the 220 service-ready response.

10. EHLO / HELO Exchange

```
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y
'smtp.request.command == "EHLO" || smtp.request.command == "HELO" \
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e
smtp.request.command -e smtp.request.parameter
```

Fra me	Time	Command	Client/se rver	Parame ter	Server response
7	Oct 5, 2009 02:06:08.2248 09000 EDT	EHLO/H ELO	Client	GP	250- xc90.websitewelco me.com Hello GP [122.162.143.157]

Advertised Capabilities Table

Capability	Observed?	Evidence
AUTH	YES	250-AUTH PLAIN LOGIN
STARTTLS	YES	250-STARTTLS
SIZE	YES	250-SIZE 52428800
Other	YES	250-PIPELINING, 250 HELP

```
response: 221 xc90.websitewelcome.com closing connection\r\n
(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y "smtp.req.command == \"EHLO\" || smtp.req.command == \"HELO\" \" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e smtp.req.command -e smtp.req.parameter
7      Oct  5, 2009 02:06:08.224809000 EDT      10.10.1.4      74.53.140.153      0      EHLO      GP
(kali@kali)-[~]
```

```
└─$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y "smtp.response.code == 250" -T fields \
    -e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.response.line
tshark: Some fields aren't valid:
    smtp.response.line
```

```
└─(kali@kali)-[~]
```

```
$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -q -z follow,tcp,ascii,0 | head -n 30
```

```
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
    181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.
```

```
9
EHLO GP
```

```
    137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP
```

```
12
AUTH LOGIN
```

```
    18
334 VXNlcm5hbWU6
```

```
30
Z3VycGFydGFwQHBhdHJpb3RzLmFu
```

```
└─(kali@kali)-[~]
```

```
$
```

o return to your computer, move the mouse pointer outside or press Ctrl+Alt.

Evidence Checkpoint 5: screenshot showing the EHLO/HELO command and the server's capability response.

11. AUTH Exchange and Offline Base64 Decoding

Example: decode one authorised training value offline

```
printf '%s' 'BASE64_VALUE' | base64 --decode
```

Alternative Python method

```
python3 -c "import base64;
```

```
print(base64.b64decode('BASE64_VALUE').decode('utf-8','replace'))"
```

Locate likely AUTH packets

```
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y
```

```
'smtp.request.command == "AUTH"' \
```

```
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e  
smtp.request.command -e smtp.request.parameter
```

Frame	AUTH mechanism	Encoded value (masked)	Decoded result (redacted)	Interpretation
10	LOGIN		AUTH LOGIN	Client initiates LOGIN authentication method
13	LOGIN	VXNlcm5hbWU=	Username	Server 334 prompt for username
14	LOGIN	Z3Vy...LmLu	[REDACTED-USERNAME]	Username base64 payload submitted by client
15	LOGIN	UGFzc3dvcmQ=	Password	Server 334 prompt for password
16	LOGIN	Z3Vy...LmLu	[REDACTED-PASSWORD]	Password base64 payload submitted by client

```
(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y "smtp.req.command == \"AUTH\" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter
10 Oct 5, 2009 02:06:08.568729000 EDT 10.10.1.4 74.53.140.153 AUTH LOGIN

(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y "smtp" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.payload | \
while read fn ft src dst payload; do \
[ -n "$payload" ] && echo "Frame $fn [$ft]:" && echo "$payload" | xxd -r -p && echo "; \
done | grep -A 8 "AUTH LOGIN"

(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y "smtp.req.command == \"AUTH\" || smtp.response.code == 334" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter -e smtp.response.line
tshark: Some fields aren't valid:
smtp.response.line

(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y "smtp.req.command == \"AUTH\" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter
10 Oct 5, 2009 02:06:08.568729000 EDT 10.10.1.4 74.53.140.153 AUTH LOGIN

(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y "smtp" -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.payload | \
while read fn ft src dst payload; do \
[ -n "$payload" ] && echo "Frame $fn [$ft]:" && echo "$payload" | xxd -r -p && echo "; \
done | grep -A 8 "AUTH LOGIN"

(kali@kali)-[~]
$ # Decode Server 334 Prompts (Username and Password requests)
echo "VXNlcm5hbWU=" | base64 -d && echo ""
echo "UGFzc3dvcmQ=" | base64 -d && echo ""

# Masked Decoding Demonstration
printf '%s' 'Z3VycFYdGFwQHBhdHJpb3RzLmLu' | base64 --decode | sed 's/./*/g'
Username
Password
*****

(kali@kali)-[~]
$
```

Evidence Checkpoint 6: screenshot of the offline decoding process with sensitive values masked/redacted.

12. MAIL FROM / RCPT TO / DATA

```
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y \
'smtp.request.command == "MAIL" || smtp.request.command == "RCPT" ||
smtp.request.command == "DATA" \
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream \
-e smtp.request.command -e smtp.request.parameter
```

Frame	Time	Command	Direction	Parameter / recipient	Response
16	Oct 5, 2009 02:06:09.614414 000 EDT	MAIL FROM	Client → Server	FROM: <[REDACTED- SENDER]@patriots.i n>	250 OK
18	Oct 5, 2009	RCPT	Client	TO: <[REDACTED-	250

	02:06:09.957250 000 EDT	TO	→ Server	RECIPIENTJ@yahoo. co.in>	Accept ed
20	Oct 5, 2009 02:06:10.320203 000 EDT	DATA	Client → Server	None	354 Enter messag e, ending with "." on a line by itself

```
(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y 'smtp.req.command == "MAIL" || smtp.req.command == "RCPT" || smtp.req.command == "DATA"' \
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e smtp.req.command -e smtp.req.parameter
16   Oct  5, 2009 02:06:09.614414000 EDT    10.10.1.4      74.53.140.153    0      MAIL    FROM: <gurpartap@patriots.in>
18   Oct  5, 2009 02:06:09.957250000 EDT    10.10.1.4      74.53.140.153    0      RCPT    TO: <raj_deol2002in@yahoo.co.in>
20   Oct  5, 2009 02:06:10.320203000 EDT    10.10.1.4      74.53.140.153    0      DATA

(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -q -z follow,tcp,ascii,0 | grep -E "MAIL FROM|RCPT TO|DATA|250|354"
250-xc90.websiteswelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP
MAIL FROM: <gurpartap@patriots.in>
250 OK
RCPT TO: <raj_deol2002in@yahoo.co.in>
250 Accepted
DATA
354 Enter message, ending with "." on a line by itself
250 OK id=IMugho-0003Dg-Un

(kali@kali)-[~]
$
```

Evidence Checkpoint 7: Insert a screenshot showing MAIL FROM, RCPT TO and DATA in sequence.

13. Follow TCP Stream — Message Reconstruction

In Wireshark, select a packet from the SMTP conversation, then use Analyze/Follow → TCP Stream. Record the stream number, direction and reconstructed SMTP dialogue. Save any permitted transcript as supporting evidence, with sensitive content redacted.

Identify stream numbers

```
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y smtp -T fields -e
tcp.stream | sort -nu
```

Inspect a selected stream

```
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y 'tcp.stream ==  
<STREAM_NUMBER>' \
```

```
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.seq -e  
tcp.ack -e tcp.len
```

Field	Observation
TCP stream	0
Client → server data	EHLO GP AUTH LOGIN [REDACTED-BASE64-USER] [REDACTED-BASE64-PASS] MAIL FROM: <[REDACTED-SENDER]@patriots.in> RCPT TO: <[REDACTED-RECIPIENT]@yahoo.co.in> DATA [MIME Message Payload & Body Contents]

	. QUIT
Server → client data	220-xc90.websitewelcome.com ESMTP Exim 4.69... 250-xc90.websitewelcome.com Hello GP... 334 VXNlcm5hbWU= 334 UGFzc3dvcmQ= 235 Authentication succeeded 250 OK 250 Accepted 354 Enter message, ending with "." on a line by itself

	250 OK id=1Mugho-0003Dg-Un 221 xc90.websitewelcome.com closing connection
Reconstruction successful	YES
Message body visible	YES

```

(kali@kali)-[~]
$ # 1. Create the evidence directory
mkdir -p ~/SBT-DF203-Lab4/evidence

# 2. Re-run the export command to save the TCP stream transcript
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -q -z follow,tcp,ascii,0 > ~/SBT-DF203-Lab4/evidence/tcp_stream_0.txt

(kali@kali)-[~]
$

```

Evidence Checkpoint 8: Follow TCP Stream screenshot showing the reconstructed SMTP exchange with sensitive data redacted.

14. Message Headers and Client Software

Within the reconstructed DATA content, locate and record the following headers where present.

Header / indicator	Observed value (redacted where necessary)	Evidence frame/stream
Date	"Gurpartap Singh" <gurpartap@patriots.in>	"Gurpartap Singh" <[REDACTED-SENDER]@patriots.in>
From	<raj_deol2002in@yahoo.co.in>	<[REDACTED-RECIPIENT]@yahoo.co.in>
To	SMTP	Standard message subject header

Subject	Mon, 5 Oct 2009 11:36:07 +0530	Client message generation timestamp (+0530 timezone offset)
Message-ID	<000301ca4581\$ef9e57f0\$cedb07d0\$@in>	Unique identifier generated by sending MUA/MTA
MIME-Version	1.0	Indicates MIME multipart format compliance
Content-Type	multipart/mixed;	Email contains multiple body parts or attachments

Search raw SMTP payload for common header strings

tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y smtp -T fields \

-e frame.number -e tcp.stream -e data.data | grep -Ei

'Date|From:|To:|Subject:|Message-ID|MIME|User-Agent|X-Mailer'

```
(kali@kali)-[~]
$ grep -i -E "Message-ID:|Date:|From:|To:|Subject:|MIME-Version:|Content-Type:" ~/SBT-DF203-Lab4/evidence/tcp_stream_0.txt
MAIL FROM: <gurpartap@patriots.in>
RCPT TO: <raj_deol2002in@yahoo.co.in>
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;
```

Evidence Checkpoint 9: screenshot showing the message headers and any client-software indicator.

15. SMTP Command / Response Timeline

Order	Time	Direction	Command / response	Code	Evidence / significance
1	Oct 5, 2009 02:06:08.22480 9000 EDT	Server → Client	220 service ready	220	Session start
2	Oct 5, 2009	Client	EHLO/HE	—	Client

	02:06:08.22480 9000 EDT	→ Server	LO		identification/capa bilities
3	Oct 5, 2009 02:06:08.22480 9000 EDT	Server → Client	Capabilitie s	250	Server advertises SIZE, PIPELINING, AUTH, STARTTLS, HELP
4	Oct 5, 2009 02:06:08.56872 9000 EDT	Client → Server	AUTH	—	Authentication
5	Oct 5, 2009 02:06:08.56872 9000 EDT	Server → Client	AUTH response	[INSER T]	235 Authentication succeeded
6	Oct 5, 2009 02:06:09.61441 4000 EDT	Client → Server	MAIL FROM	—	Envelope sender
7	Oct 5, 2009 02:06:09.61441 4000 EDT	Server → Client	MAIL response	250	250 Accepted
8	Oct 5, 2009 02:06:09.95725 0000 EDT	Client → Server	RCPT TO	—	Envelope recipient
9	Oct 5, 2009 02:06:09.95725 0000 EDT	Server → Client	RCPT response	250	250 Accepted
10	Oct 5, 2009 02:06:10.32020 3000 EDT	Client → Server	DATA	—	Message transfer
11	Oct 5, 2009 02:06:10.32020 3000 EDT	Client → Server	Message headers/b ody	—	Payload transmission ending with . on a line by itself
12	Oct 5, 2009 02:06:10.32020 3000 EDT	Server → Client	DATA response	250	Delivery queued (250 OK id=1Mugho-0003Dg-

					Un)
13	Oct 5, 2009 02:06:10.32020 3000 EDT	Client → Server	QUIT	—	Session termination

```

(kali@kali)~$ tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -q -z follow,tcp,ascii,0 | grep -E "(220|EHLO|250|AUTH|235|MAIL|RCPT|DATA|354|QUIT|221)"
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.
EHLO GP
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP
AUTH LOGIN
235 Authentication succeeded
MAIL FROM: <gurpartap@patriots.in>
250 OK
RCPT TO: <raj_deol2002in@yahoo.co.in>
250 Accepted
DATA
354 Enter message, ending with "." on a line by itself
250 OK id=1Mugho-0003Dg-Un
QUIT
221 xc90.websitewelcome.com closing connection
(kali@kali)~$

```

Evidence Checkpoint 10: screenshot or consolidated evidence table supporting the final SMTP timeline.

16. STARTTLS / TLS Assessment

Search for STARTTLS

tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y 'frame contains "STARTTLS" \

-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e data.data

Identify TLS packets

tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y tls -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e
tls.record.content_type

Item	Observed result
STARTTLS command present	NO
STARTTLS response	N/A (Capability advertised by

	server, but command not issued by client)
TLS handshake present	NO
TLS version	N/A
Certificate information visible	NOT APPLICABLE
SMTP commands after TLS readable?	YES (<i>Session was entirely in cleartext</i>)
Message content after TLS readable?	YES (<i>Session was entirely in cleartext</i>)

```

(kali@kali)-[~]
$ # 1. Check for STARTTLS command usage
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y 'frame contains "STARTTLS"' -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream
# 2. Check for TLS packets
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y tls -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e tls.record.content_type
9      Oct  5, 2009 02:06:08.566183000 EDT      74.53.140.153      10.10.1.4      0
(kali@kali)-[~]
$

```

Evidence Checkpoint 11: screenshot supporting the STARTTLS/TLS conclusion.

17. QUIT and Session Termination

```
tshark -r ~/SBT-DF203-Lab4/working/smtp.pcap -Y
```

```
'smtp.request.command == "QUIT"' \
```

```
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e
smtp.request.command
```

Frame	Time	Direction	Command	Server response	Interpretation
23	Oct 5, 2009 02:06:10.32020 3000 EDT	Client → Server	QUIT	221 xc90.websitewelcome.com closing connection	Session termination

```

(kali@kali)-[~]
$ # 1. Check for STARTTLS command usage
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y 'frame contains "STARTTLS"' -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream

# 2. Check for TLS packets
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y tls -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e tls.record.content_type
9   Oct  5, 2009 02:06:08.566183000 EDT    74.53.140.153    10.10.1.4    0

(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y 'smtp.req.command = "QUIT"' \
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e smtp.req.command
54   Oct  5, 2009 02:06:14.763825000 EDT    10.10.1.4      74.53.140.153    0    QUIT

(kali@kali)-[~]
$ tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -q -z follow,tcp,ascii,0 | grep -E "QUIT|221"
QUIT
221 xc90.websiteswelcome.com closing connection

(kali@kali)-[~]
$

```

18. Quantitative Findings

Measure	Observed value
Total packets in PCAP	60
SMTP-related packets	32
TCP conversations	1
SMTP TCP streams	1
Distinct client Ips	1 (10.10.1.4)
Distinct server Ips	1 (74.53.140.153)
Email transactions identified	1
STARTTLS sessions	0
Plaintext SMTP sessions	1

```
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -q -z io,phs
```

```
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y smtp -T fields -e
tcp.stream | sort -nu | wc -l
```

```

(kali@kali)-[~]
$ # Total packets in PCAP
capinfos -c ~/SBT-DF203-Lab4/working/sntp.pcap | grep "Number of packets"

# SMTP-related packets
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y smtp | wc -l

# Total TCP conversations
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -q -z conv,tcp | grep -E "↔" | wc -l

# SMTP TCP streams
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y smtp -T fields -e tcp.stream | sort -nu | wc -l

# Distinct client IPs and server IPs
tshark -r ~/SBT-DF203-Lab4/working/sntp.pcap -Y smtp -T fields -e ip.src | sort -u
Number of packets: 60
32
1
1
10.10.1.4
192.168.1.1,10.10.1.4
74.53.140.153

(kali@kali)-[~]
$

```

19. Forensic Findings

Finding	Evidence	Assessment
Email exchange time	Oct 5, 2009 02:06:08.224809000 EDT - Oct 5, 2009 02:06:10.320203000 EDT	Transmission completed in approximately 2.095 seconds.
SMTP client	IP: 10.10.1.4 / Software: EHLO GP	Private network address acting as Mail User Agent / Client.
SMTP server	IP: 74.53.140.153 / Hostname: xc90.websitewelcome.com	Remote ESMTP Exim 4.69 mail server.
Envelope sender	MAIL FROM: <[REDACTED-SENDER]@patriots.in>	Validated sender domain matching message headers.
Envelope recipient	RCPT TO: <[REDACTED-RECIPIENT]@yahoo.co.in>	Validated target recipient address accepted by server (250 Accepted).
Subject	Subject: SMTP	Unencrypted message subject line.
Authentication observed	YES	Cleartext AUTH LOGIN method executed; credentials Base64 decoded and redacted.
STARTTLS/TLS	NO	Unencrypted message subject line.
Message reconstruction	COMPLE	Cleartext AUTH LOGIN method executed; credentials Base64 decoded and redacted.

20. Evidential Limitations

- A packet capture is evidence of observed network traffic; it does not by itself prove who physically operated the client system.
- IP addresses identify network endpoints, not necessarily individual persons.
- MAC addresses may identify local interface addresses within the captured segment but do not establish ownership of a person.
- Base64 decoding reveals encoded data but does not provide cryptographic security.
- If STARTTLS/TLS was negotiated, later SMTP commands and message content may be unavailable for plaintext reconstruction.
- A missing packet, truncated capture or asymmetric capture point can prevent complete reconstruction.
- Redaction in this report intentionally hides sensitive credentials and personal message content; the preserved authorised evidence remains the source for verification.

21. Conclusion

The SMTP forensic investigation demonstrated a repeatable process for preserving packet captures, establishing evidence integrity via SHA-256 hashing, identifying client/server endpoints, parsing text-based SMTP commands and responses, decoding Base64 authentication structures offline, extracting message headers, and assessing protocol security states (STARTTLS/TLS). All conclusions in this report are based strictly on captured network packet evidence.